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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PNG 520 — Past Questions & Materials
by Renance

Course: PNG 520

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 500

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Paper 1 of 2 — PNG 520 NATURAL GAS PROCESSING AND UNIT OPERATIONS TEST QUESTIONS 2022

Level 500 • Second Semester • PastQuestion

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2022 QUIZ

PNG 520 - NATURAL GAS PROCESSING & UNIT OPERATION
Matriculation No. 10-1100006017

#9. What is a thermodynamic process? *This is the branch of physics concerned with heat and temperature and their relation to energy and work.*

#10. The gas in a cylinder in figure 1 is at room temperature ($T = 293\text{ K}$), when it was heated. The gas expanded and moving the piston originally confining the gas up a distance d (see fig. 2). If the volume occupied by the gas triples when the gas heated, how much work has the gas done?

Original gas volume $= 0.075\text{ m}^3$, Piston weight $[W] = 200\text{ N}$, Piston area $= 0.56\text{ m}^2$.

$P_g A = P_{atm} A + W$
 $P_g = \frac{P_{atm} A + W}{A} = \frac{1013 \times 10^3 + 200}{0.56} = 101057.14$

Since Vol is $\times 3 =$
 $T_2 = 2 \cdot T_1 = 2 \cdot 293\text{ K} = 586\text{ K}$
 $P_2 V_2 = P_1 V_1$
 $P_2 = \frac{P_1 V_1}{V_2} = \frac{101057.14 \times 0.075}{0.225} = 33685.71$

Fig. 1

Fig. 2

#11. What is the difference between Internal Energy and Thermal Energy?
Internal energy is the sum of the kinetic energy and potential energy of the constituent particles in a system, while Thermal energy is the average kinetic energy of the constituent particles due to their motion.

#12. For a particular pressure the evaporation of liquid will take place at all temperatures
 a. FALSE
 b. TRUE

#13. For a particular pressure the boiling of liquid will take place at all temperatures
 a. FALSE
 b. TRUE

#14. Conduction is the transfer of heat energy by the direct interactions of atoms or molecules of a material and it does not occur in liquids and gas.
 a. FALSE
 b. TRUE

#15. Define reservoir fluid separation process. *It is a process that involves the separation of reservoir fluid into its component phases.*

#16. Using the mechanisms of separation involved and in a bullet format, describe the separation process in the secondary separation section of a typical oil and gas production separator.

- i - Gas creams out of the oil
- ii - Oil settle down from the gas
- iii - Water will settle down to the bottom
- iv - Oil cream out of the water

$\Rightarrow$ Due to Gravity Difference

PNG 520 – NATURAL GAS PROCESSING & UNIT OPERATION

2022 QUIZ Matriculation No. 2012020202

22. Sketch and label a typical shell and tube heat exchanger with hot gas passing through the shell side and cold water passing through the 6 tubes of the tube side.

23. Label the diagram below

24. Sketch and label a typical phase diagram of hydrocarbon system

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A. Oil Well

B. Separator

C. Separator

D. Separator

E. Separator

F. Separator

G. Separator

H. Separator

I. Separator

J. Separator

K. Separator

L. Separator

M. Separator

N. Separator

O. Separator

P. Separator

Q. Separator

R. Separator

S. Separator

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Matriculation No. 02150001207

2022 QUIZ

#17. The transfer of heat through conduction does not involve movement of molecules.
A. FALSE
B. TRUE

#18. Define and state the Equilibrium Calculation Equation. This is the calculation equation used to determine the quantities and composition of gas and liquid mixture in equilibrium.
$$\sum x_i = \frac{\sum y_i}{1 + K_i(y_i)}$$

#19. What does each parameter of the Equilibrium Calculation Equation represent?
 x_i = mole fraction of component i in liquid
 y_i = mole fraction of component i in gas
 n_g = number of mole in the gaseous phase
 n_L = number of mole in liquid phase
 K_i = equilibrium constant
 z_i = mole fraction of i in mixture

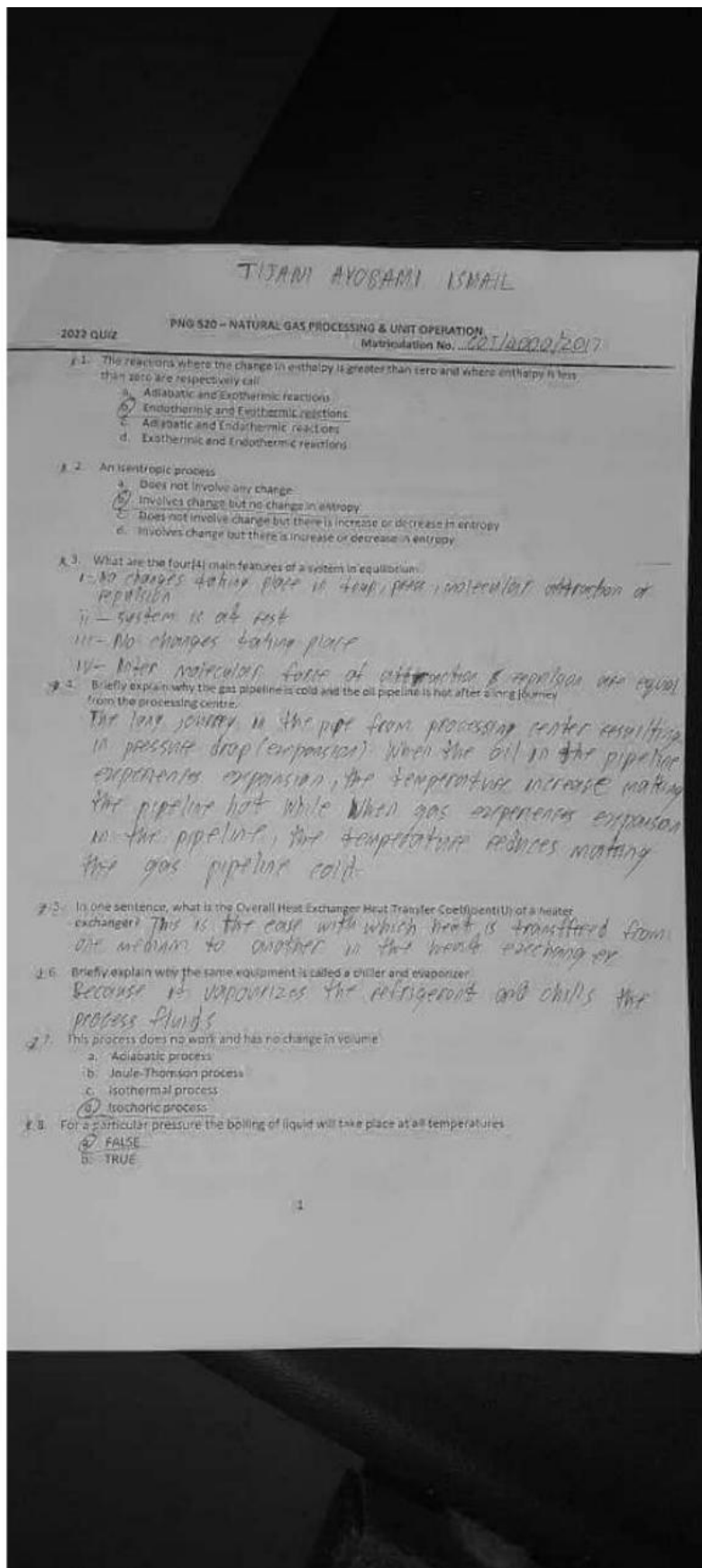
#20. Label the mechanical compression refrigeration cycle below.

A. Evaporator
B. Condenser
C. Compressor
D. Expansion valve
E. Liquid vapor
F. Vapor

#21. Heat is being transferred at a rate 2.4×10^5 joules/(sec.m²) across a 1.5mm thick metal with thermal conductivity constant of 240 J/m.s.K. What will be the final temperature of the 1.5mm thick metal that was originally 75°C?
 $K = 240 \text{ J/m.s.K}$; $\Delta L = 1.5 \text{ mm} = 0.0015 \text{ m}$; $Q = 2.4 \times 10^5 \text{ joules/(sec.m}^2)$ and $T_1 = 75^\circ\text{C}$; $T_2 = ?$

$$Q = \frac{K \Delta T}{\Delta L} \quad \Delta T = \frac{Q \Delta L}{K} = \frac{2.4 \times 10^5 \times 0.0015}{240} = 15^\circ\text{C}$$

$$T_2 = T_1 + \Delta T = 75^\circ\text{C} + 15^\circ\text{C} = 90^\circ\text{C}$$



Paper 2 of 2 — 2018 PNG 520 NATURAL GAS PROCESSING AND UNIT OPERATION QUIZ

Level 500 • Second Semester • PastQuestion

DEPARTMENT OF PETROLEUM ENGINEERING

PNG 520: Natural Gas Processing and Unit Operations

Duration -- 60 mins --- 2018 PNG 520 Quiz 1 Mat. No.

1. Sketch and label a block process flow diagram of associated gas(AG)

processing plant to produced sale natural gas and LPG – Label with functional processing names not generic names. Sales

Separator Natural G

AG Gas Refrigeration

System

Compression C2+

Station

LPG

NGL

C3+

Debutanizer

Deethanizer

2. Define thermodynamics

Thermodynamics is a branch of physics concerned with heat and

Gasoline (C5+)

temperature and their relation to energy and

work

It is the branch of physical science that deals with the relationship

between heat and other forms of energy (such as mechanical,

electrical or chemical energy)

3. Mention three main areas of application of thermodynamics in natural gas processing

a. Heat Transfer

b. LGP production

c. NGL production

d. LNG production

e. Joule Thomson Expansion

f. Refrigeration processes

g. Throttling valves, etc

4. What process of heat transfer is perspiration?

a. Convection

- b. Conversion
- c. Transpiration
- d. Evaporation

5. Define mass transfer and give two examples of processes of mass transfer

Definition: This is the transfer of mass into or out of a substance or transfer of a chemical compound from one phase to another

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PNG 520: Natural Gas Processing and Unit Operations

Examples:

- a. Evaporation of liquid to gas
- b. Diffusion of solute from high concentration to low concentration
- c. Absorption of one or more compounds from gas phase to a liquid phase
- TEG dehydration

6. Calculate the water concentration of the rich glycol leaving the natural gas dehydrating contactor that was fed with re-concentrated glycol entering at the rate of 1500 kmol/h with $x_A = 0.001$. The natural gas entering into the contactor flows at a rate of 500 kmol/h with $y_A = 0.3$ and leaving the contactor with $y_A = 0.01$. Assume equilibrium relationship is $y_A = 2.8 x_A$.

- a. Calculate the actual flow rate of the glycol without solute conc.
- b. Calculate the actual flow rate of the natural gas without solute conc.
- c. Calculate the concentration of water in the rich glycol

Now,

7. Absolute zero is

- a. A point at which the kinetic energy of a system is equal to the potential energy
- b. A point at which the fundamental particles of nature have NO vibrational motion
- c. Zero degree Fahrenheit
- d. A point at which the fundamental particles of nature have minimal vibrational motion

8. Water vapour turning to ice is

- a. Sublimation
- b. Evaporation
- c. Deposition
- d. Condensation

9. Give three(3) applications of heat transfer in natural gas processing

a. Reconcentration of Rich Glycol in dehydration

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PNG 520: Natural Gas Processing and Unit Operations

b. Breaking out of absorbed H₂S, CO₂ from amine in sweetening

c. Condensation of NGL from natural gas

d. Liquefaction of natural gas

e. Heating off absorbed water from solid desiccant

f. Thermal control to achieve quality temperature for sales gas and equipment requiring certain temperature for processing operation

10. Define phase equilibrium in absorption dehydration process

This is the point during the absorption process when the rate at which absorbed substance entering the liquid phase exactly equals that at which it leaves the gas phase. At this point, the net mass transfer molecules travelling back and forth from one phase to another is zero

11. Briefly describe the difference between the vaporizer and evaporator(chiller)

A vaporizer adds heat to liquid to change it to gaseous phase while in an evaporator process fluid is cooled through heat removal by a flowing pool of refrigerant

12. What is the difference between phase behaviour and phase diagram?

Phase behaviour is the changes in phase exhibited by the natural gas when subjected to different temperature-pressure conditions while the phase diagram is the pressure temperature diagrammatic representation of the various temperature-pressure conditions at which the natural gas exist in different phases

13. Briefly describe the process of flash evaporation leading to equilibrium condition n ; y

g ; j

Flash

Evaporation n_L ; x_j

P & T

n ; Z_j Equilibrium

When liquid is subjected to a sudden pressure change beyond its bubble pressure, a flash evaporation of the substance vapour occurs to form a gaseous phase. If the new set of pressure and temperature is maintained, the gaseous and liquid will attain

equilibrium condition whereby equilibrium equation can apply to determine their quantities of the phases and their respective components and mole fractions

14. The equilibrium calculation equations are applied to phase equilibrium condition when the number of molecules leaving one phase to the other phase are equal to the ones coming in.

a. TRUE

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b. FALSE

15. Calculate the rate of heat transfer across 1.5mm thick metal with thermal conductivity constant of 240 J/m s K when heat was used to raise its temperature from 60 °C to 75 °C.

$K = 240 \text{ J/m s K}$; $\Delta L = 1.5\text{mm} = 0.0015\text{m}$; and $\Delta T = 75 - 60 = 15$

Now

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